Multimodal Tomato Leaf Disease Detection

Fusing Computer Vision, NLP, and Environmental Sensors

An AI assistant that diagnoses tomato diseases by combining visual analysis, symptom reports, and weather data—mimicking how expert agronomists think.

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The Challenge: Why Vision Alone Fails

The Problem

Standard agricultural AI relies solely on photographs, but different diseases often appear identical on camera. Early Blight and Septoria Leaf Spot can look nearly indistinguishable visually, leading to misdiagnosis and incorrect treatment.

The Human Expert Approach

Real agronomists don't just look at leaves—they ask about weather patterns, symptom history, and growing conditions to make informed diagnoses.

The Solution: A Multimodal Approach

Computer Vision

High-resolution imagery processed through deep convolutional networks to extract visual disease patterns and leaf characteristics.

Natural Language Processing

Farmers describe symptoms in plain English, which the AI interprets to understand context and progression.

Environmental Data

Live temperature and humidity data provide critical epidemiological context for fungal disease prediction.

By fusing these three data streams, the AI makes context-aware diagnoses that match expert agronomist accuracy.

Synthesizing the Dataset

01

Base Image Collection

5,000+ tomato leaf images spanning 10 health states from PlantVillage dataset

02

Weather Simulation

Mathematically generated temperature and humidity parameters matched to real-world epidemiology

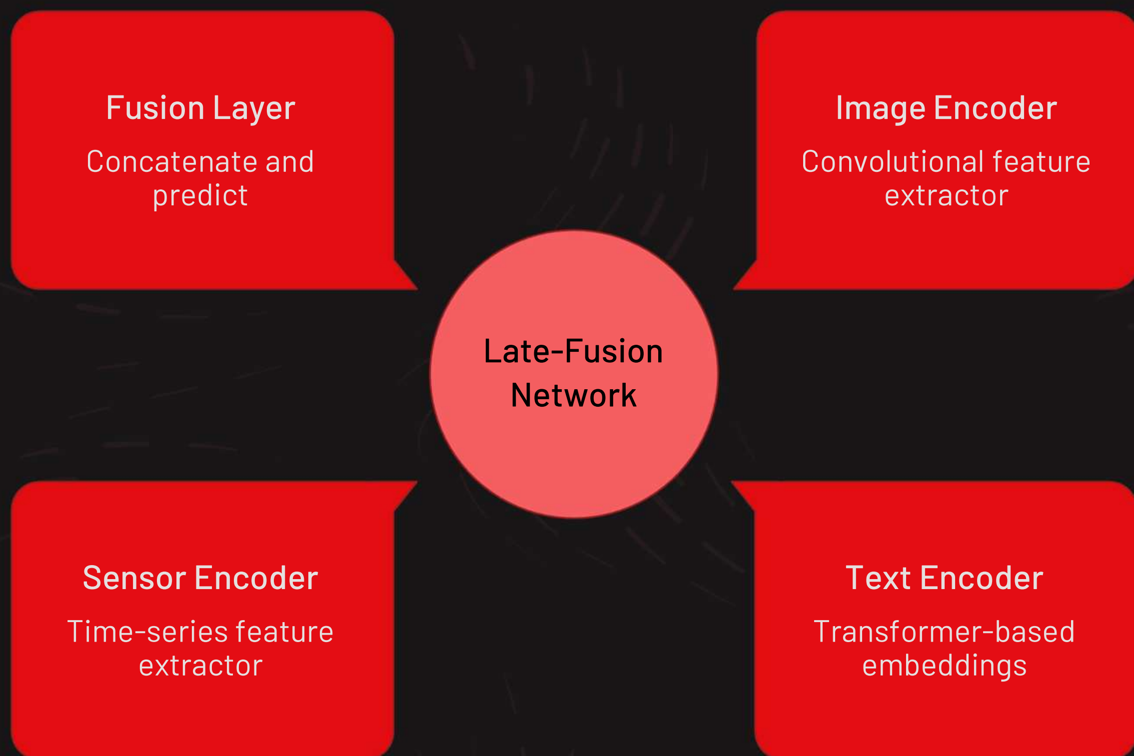
03

Symptom Text Generation

Natural language descriptions aligned with disease progression and environmental conditions

No public datasets perfectly align images, farmer text, and weather data—so I synthesized my own training set.

System Architecture: Late-Fusion Network



The architecture uses a **late-fusion strategy**: each data modality is processed independently by dedicated neural networks before [illegible] their learned representations at the final decision layer.

Three Encoders: Vision, NLP, Sensors

Vision Encoder

ResNet50 (pre-trained)

Extracts 2048-dimensional visual feature vectors by removing final classification layer

NLP Encoder

OpenAI CLIP transformer

Processes semantic text into 512-dimensional vectors capturing symptom meaning

Sensor Encoder

Multi-Layer Perceptron

Scales temperature and humidity into 128-dimensional vectors compatible with image features

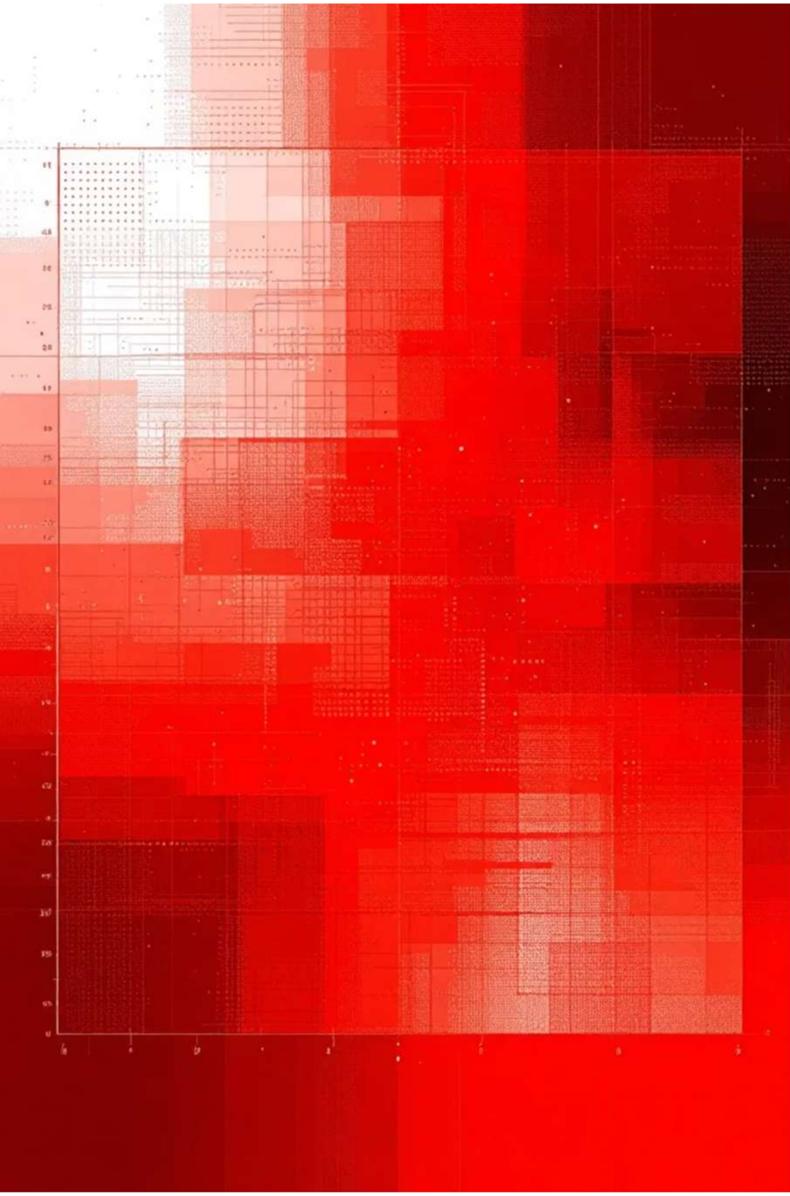
The Fusion Layer: Making the Final Diagnosis

Architecture Details

- Input vectors concatenated into unified 2688-dimensional array
- Dense neural network with Batch Normalization and Dropout (30%)
- Softmax output layer for 10 disease classes
- Trained with cross-entropy loss

What the AI Learns

The fusion layer learns relationships between modalities—for example, that ambiguous images with 95% humidity strongly indicate Late Blight.



Model Performance Metrics

96%

Overall Accuracy

Correctly classified 96% of test samples across all 10 disease categories

94%

F1-Score

Balanced precision and recall demonstrating robust classification

80/20

Train/Validation Split

Ensured model generalizes to unseen data

The multimodal approach significantly reduced false positives between visually similar fungal infections by leveraging environmental context.

Chainlit UI: User-Friendly Deployment

Image Upload

Drag-and-drop interface for high-resolution tomato leaf photos

Weather Sliders

Real-time temperature and humidity adjustment controls

Text Description

Optional symptom details in plain English for enhanced context

PyTorch model deployed via Chainlit creates an intuitive chatbot that processes multimodal inputs instantly and delivers actionable diagnoses.

Actionable Output & Future Directions

What Farmers Receive

- Confidence score for diagnosis
- Plain-English disease explanation
- Environmental contributing factors
- Immediate mitigation steps
- Specific fungicide recommendations

Next Steps

- **IoT Integration**

  Direct API connection to live greenhouse sensors

- **Edge AI Deployment**

  Model quantization for offline mobile use

- **Expanded Crop Coverage**

  Extension to pepper, potato, and other solanaceous crops

Multimodal fusion represents the next logical step in agricultural AI—moving beyond vision-only systems to context-aware assistants that think like human experts.

THANK YOU!!!